

MINI - PROJECT (23ET5PWMPR)
based report
ON
“LifeLine Helmet: Alcohol-Free Ignition & Emergency SOS
Alert”

Submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING
IN
ELECTRONICS AND TELECOMMUNICATION ENGINEERING



VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI

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Department of Electronics and Telecommunication Engineering

B.M.S COLLEGE OF ENGINEERING

(Autonomous Institution under VTU, Belagavi)

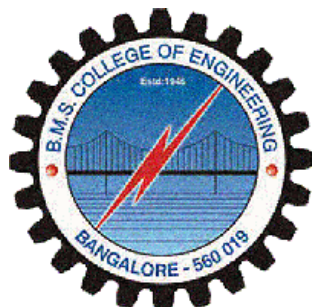
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BONAFIDE CERTIFICATE

This is to certify that the Mini project (23ET6PWMPR) entitled “**LifeLine Helmet: Alcohol-Free Ignition & Emergency SOS Alerts**” is submitted by **Hasheer Ahmed (USN:1BM23ET018)**, **M D Zameer Hussain (USN:1BM23ET027)**, **Muhammad Furqan (USN:1BM23ET033)** and **Vinay B Subramanya (USN:1BM23ET057)**, in partial fulfillment for the award of degree of Bachelor of Engineering in Electronics and Telecommunication Engineering during the academic year 2024-2025.

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We would like to thank the members of the committee for all their dedication, hard work, and teamwork.

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Finally, we would like to thank BMS College of Engineering for the resources provided, which made this prototype possible.

ABSTRACT

This study addresses motorcycle safety, which remains a concern due to the substantial share of two-wheelers in traffic and the presence of continued risk factors like alcohol-impaired riding and inconsistent helmet use. While traditional approaches create awareness through campaigns and enforcement that contribute to safety, they do not prevent unsafe vehicle operation from the moment of startup.

The system, Smart Helmet–Bike, will align safety logic with the behavior of a rider. More specifically, motorized two-wheelers would be authorized to start only when a rider is wearing a helmet and presents a sober breath reading. Accordingly, the proposed solution consists of an alcohol sensor embedded in the helmet, a mechanism to detect the wearing of a helmet, and a secure wireless link to a relay-based interlock integrated into the bike's ignition circuit.

Moreover, the system enables accident detection automatically. Severe, sudden deviation based on detection by an accelerometer/gyroscope triggers an instant emergency SOS alert. By utilizing low-cost, widely available components such as the ESP32, MQ-3 sensor, and FSR sensor, this design has the potential to provide a practical, scalable, and affordable solution to reduce accidents involving intoxication or failure to wear a helmet.

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CHAPTER 1: INTRODUCTION

1.1. Overview

This research addresses motorcycle safety, a critical issue in contemporary transportation. Traditional interventions have often failed to preemptively mitigate unsafe riding behaviors. The project idea proposes the development of the LifeLine Helmet, a system designed to incorporate Alcohol-Free Ignition control and Emergency SOS Alerts. The system is configured to allow the ignition of the motorcycle only when the rider is sober and wearing the helmet.

1.2. Problem statement

Two-wheeler accidents are a major cause of road deaths in India. In fact, most such accidents result either from driving under the influence of alcohol or not wearing a helmet. Traditional safety measures involving fines on road traffic and awareness campaigns cannot ensure compliance by the riders. This indicates an obvious need for a technological solution that makes compulsive wear of a helmet, while attempting to start a motorcycle, and interlocks starting up of motorcycles by intoxicated riders.

1.3. Proposed solution

The present project addresses this gap by developing a Smart Helmet system that integrates alcohol detection, verification of helmet-wearing, and ignition control. The core objectives are:

- **Integrated Detection:** Alcohol consumption using MQ-3 sensor and helmet wearing using the FSR (force sensitive resistor).
- **Ignition Control:** The bike should only start up when the rider is sober and with their helmet on.
- **Emergency Alert System:** Automatic accident detection featuring instantaneous emergency SOS alerts.

CHAPTER 2: LITERATURE SURVEY

Several research efforts have explored smart helmet systems in the interest of improving rider safety. The following summarizes the relevant literature reviewed for this project.

[1] ResearchGate, “Smart Helmet with Alcohol Sensing and Bike Authentication for Riders,” 2022.

Focuses on the authentication aspect between the helmet and the bike unit.

- **Key Finding:** Highlights the importance of wireless link encryption between helmet and bike for preventing tampering.

[2] IARJSET, “Smart Helmet Review with Accident Detection and Communication,” 2024.

Focus on systems that detect accidents in addition to preventing them.

- **Key Finding:** Demonstrates the integration of communication modules for post-accident safety (SOS alerts).

[3] RJWAVE Journal, “Helmet with Alcohol Detection and GPS/GSM for Road Safety,” 2025.

Investigates the inclusion of GPS for location tracking.

- **Key Finding:** GPS integration improves the effectiveness of emergency alerts by providing precise location data.

[4] IJRTI, “Smart Helmet with Accident Detection and Safety Features,” 2025. A comprehensive study on combining multiple safety features into a single wearable unit.

- **Key Finding:** Confirms that multi-sensor systems (Alcohol + Impact + Wear detection) are the future of rider safety.

[5] IJRPR, “Smart Helmet with Alcohol Detection and Ignition Control,” 2022.

his paper discusses the foundational concept of linking helmet sensors to ignition control.

- **Key Finding:** Established the feasibility of using alcohol sensors to interrupt ignition circuits.

CHAPTER 3: COMPONENTS

3.1 Microcontroller (ESP32):

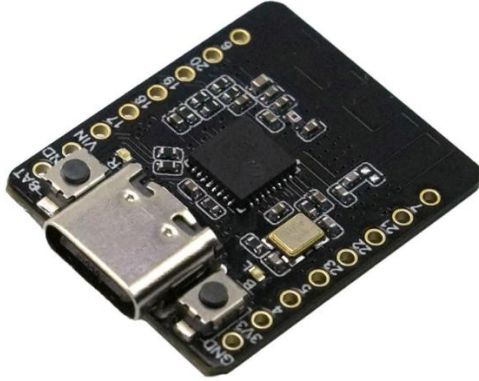


Fig 3.1: DFRobot ESP32 C6



Fig 3.2: ESP32 Dev Module

For the Helmet Unit design, the DFRobot ESP32 C6 is used as the main microcontroller unit due to its powerful processing abilities and embedded wireless communication capabilities via Bluetooth and Wi-Fi. This will be beneficial in real-time, effective acquisition and processing of data from all helmet-mounted sensors. Such processed information is then wirelessly linked to the secondary ESP32 Dev Module in the Bike Unit, which is the focal point for logging, displaying, and transmitting the information to external monitoring sources for the coordination of both the rider and the vehicle.

3.2 MQ-3 Alcohol Sensor:

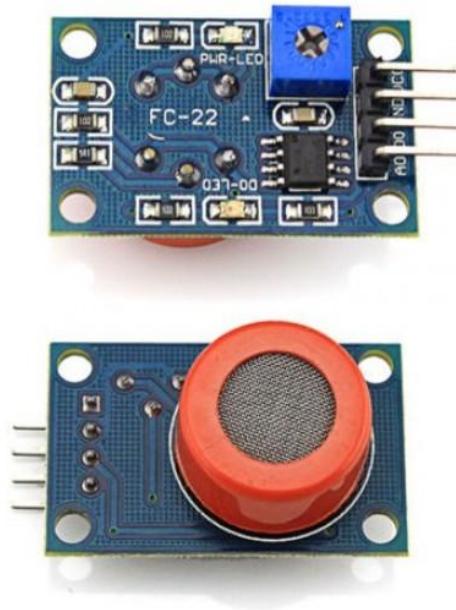


Fig 3.3: MQ-3 Sensor

The MQ-3 sensor provides an analog output proportional to the alcohol concentration in the rider's breath. This necessitates calibration for environmental stability.

3.3 Force Sensitive Resistor:

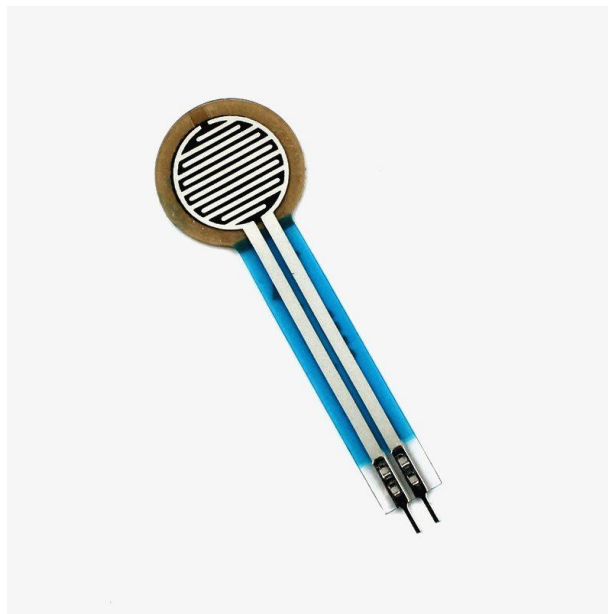


Fig 3.4: Force sensitive resistor

Inside the inner padding, there is an embedded Force Sensitive Resistor that verifies wear of the helmet. The FSR is a robust polymer thick film device whose resistance decreases with applied force. The contact pressure of the head is detected, which ensures that the helmet is properly secured and circumvention cannot be made by merely putting an object inside the helmet.

3.4 Relay Module:

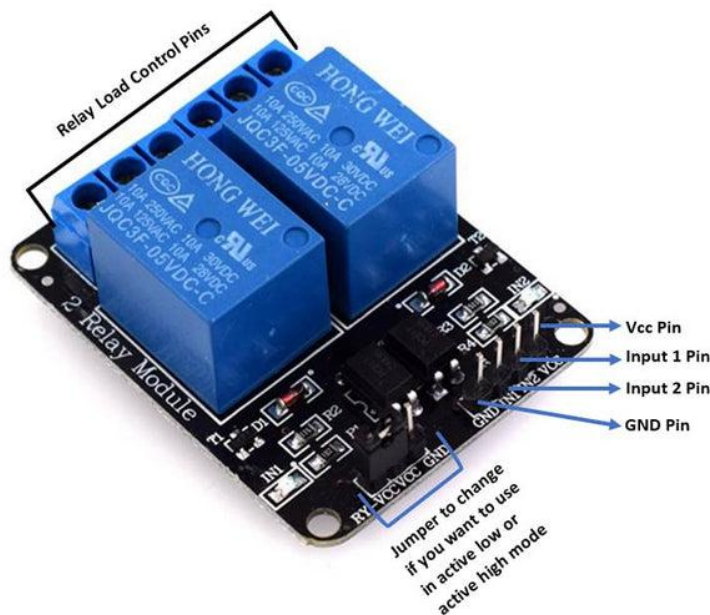


Fig 3.5: Relay Module

Situated in the Bike Unit, the 5V Relay Module serves as an actuator for the ignition circuit. It only closes the circuit (allowing engine start) when valid signals ('1') are received from the helmet unit.

3.5 MPU-6050 & GPS:



Fig 3.6: MPU-6050

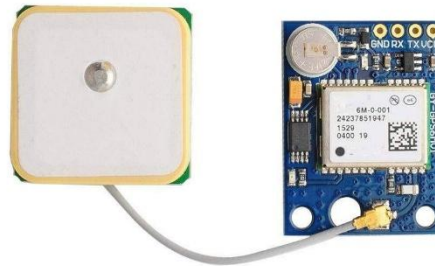


Fig 3.7: Neo-6M GPS

- **MPU-6050:** This is an accelerometer and gyroscope module. It serves to detect great deviations or impact signals-for example, indicating a crash.
- **Neo-6M GPS:** supplies location data to transmit during an SOS alert.

3.6 16x2 I2C LCD display:



Fig 3.8: 16x2 I2C LCD display

This display is used to show the ignition status and accident detected.

3.7 Power Supply:



Fig 3.8: Battery (3.7v)

The helmet unit contains a rechargeable Li-ion battery to ensure portability and extended operation.

CHAPTER 4: METHODOLOGY

4.1 System Architecture:

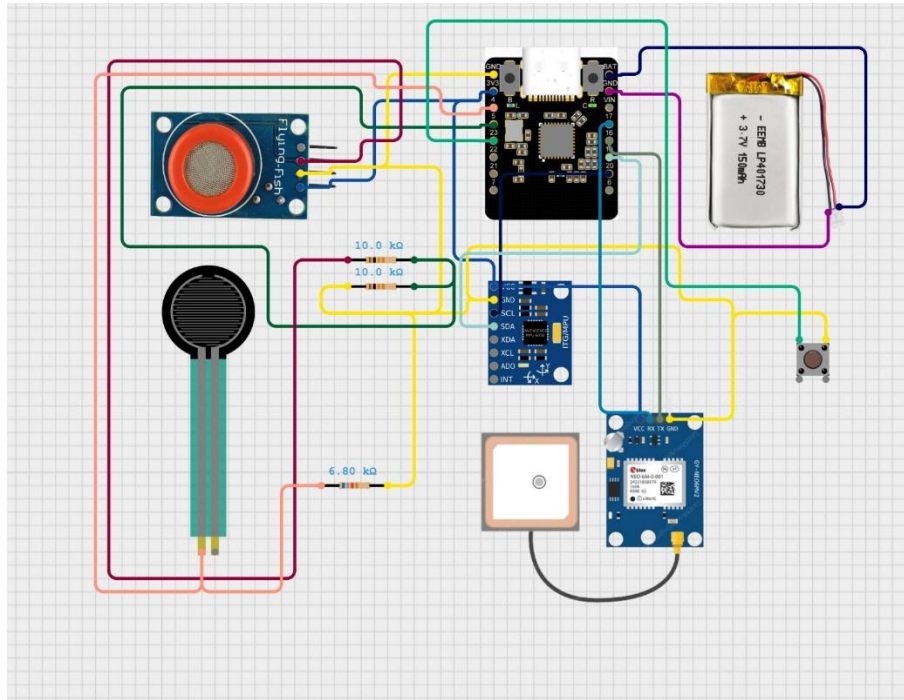


Fig 4.1: Helmet Module

Helmet Unit (Data Acquisition):

The Helmet Unit is the central sensing hub worn by the rider; it focuses on a compact microcontroller- ESP32, continuously monitoring inputs from multiple safety sensors:

- **The Sensors:** MQ-3 analyzes the breath for traces of alcohol; the FSR inserted in the padding detects head pressure to confirm helmet use.
- **Safety Monitors:** Simultaneously, the MPU-6050 (accelerometer/gyroscope) and Neo-6M GPS module work in the background. They are responsible for detecting sudden impact forces indicative of a crash and pinpointing the location for emergency alerts.
- **Power:** The entire wearable unit is powered by a compact, rechargeable Li-Po battery, keeping it lightweight and portable

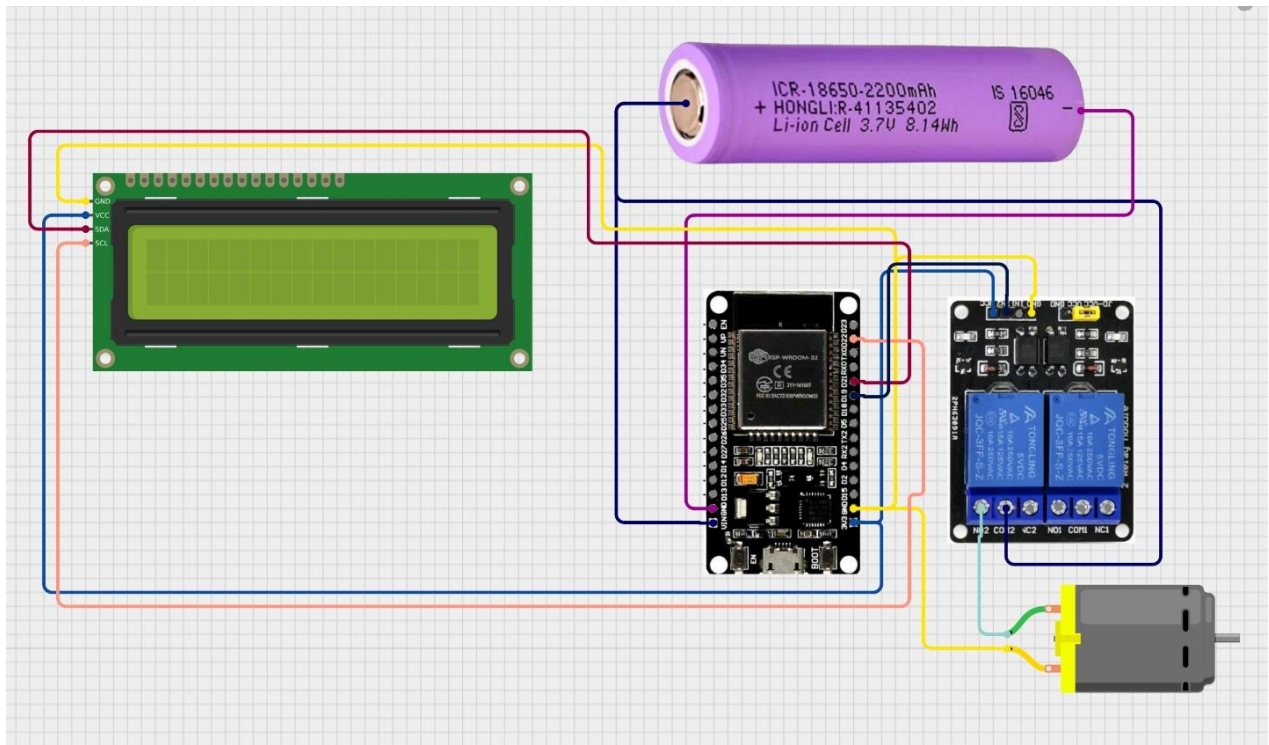


Fig 4.2: Bike Module

Bike Unit (Control & Actuation): The Bike Unit acts as the "receiver" and physical controller installed on the motorcycle:

- **The Controller:** An ESP32 board serves as the control brain; it receives wireless safety clearance signals from the helmet.
- **The "Electronic Key":** The Relay Module would be the intelligent switch for the ignition system, leading to the DC motor/engine starter. The relay would close the circuit only for engine start upon confirmation of sobriety and proper protection by the helmet.
- **User Interface:** An LCD Display-I2C serves to provide immediate feedback. For example, it would display messages like "System Ready" or "Alcohol Detected."
- **Power:** A robust 18650 Li-ion battery powers the relay coils, display, and microcontroller.

4.2 Workflow:

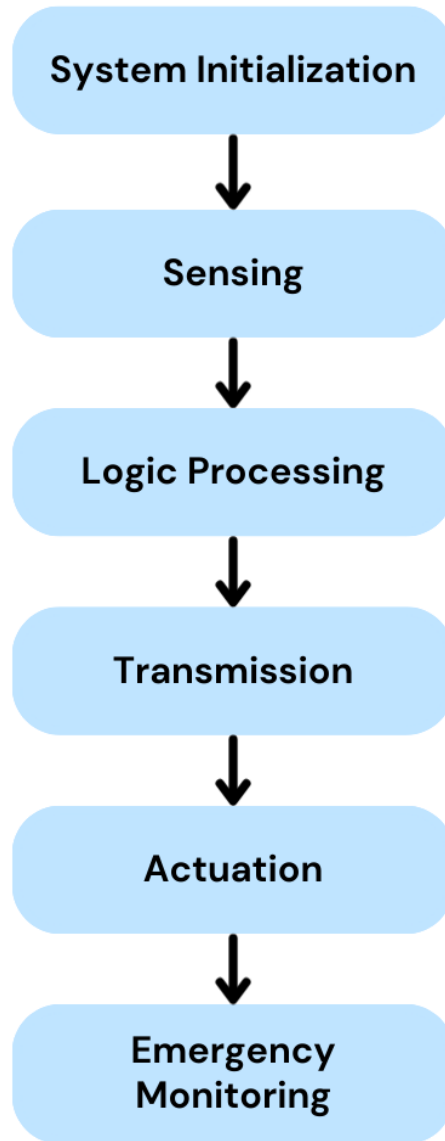


Fig 4.3 Workflow

4.2.1 System Initialization:

The microcontrollers power up, and sensors (MQ-3, MPU-6050) undergo calibration.

4.2.2 Sensing:

- The FSR sensor checks if the helmet is worn.
- The MQ-3 sensor analyzes breath for alcohol.

4.2.3 Logic Processing:

- **Condition A:** Helmet worn AND alcohol not detected.
- **Condition B:** Helmet not worn OR alcohol detected.

4.2.4 Transmission:

The signal transmitted by the Helmet Unit via a wireless link to the Bike Unit is '1' for Condition A and '0' for Condition B.

4.2.5 Actuation:

- If '1' is received, the Relay gets activated, and the ignition circuit closes to let the bike get started.
- If '0' is received: Relay stays open - Startup not allowed.

4.2.6 Emergency Monitoring:

Finally, it is important to note that concurrently, the MPU-6050 monitors for crash events. Therefore, a crash detection triggers an automatic emergency text notification.

CHAPTER 5: IMPLEMENTATION

The LifeLine Helmet: Alcohol-Free Ignition & Emergency SOS Alert system was realized in a practical hardware-software module integration. The resulting prototype contains two major components: the Helmet Unit and the Bike Unit. These two units work together to enforce safety and send out emergency notifications.

Hardware Implementation

The Helmet Unit uses an ESP32-C6 microcontroller as the central processing element, and interfaces with:

- MQ-3 Alcohol Sensor located proximal to the rider's oral region to detect alcohol in exhaled breath.
- FSR installed within the inner padding of the helmet to confirm correct wearing of the helmet.
- MPU-6050 accelerometer and gyroscope to detect crashes via monitoring sudden or anomalous movements.
- A lightweight lithium-ion (Li-ion) battery to deliver portable power.

All the components are mounted on a compact PCB and securely housed inside the helmet without compromising comfort.

The Bike Unit uses another ESP32 module and:

- Receives wireless safety status from the helmet
- Controls a 5V Relay Module placed in series with the ignition switch, allowing ignition only upon receipt of a valid safety signal.
- The I2C 16x2 LCD display shows various rider notifications like "Helmet Not Detected," "Alcohol Detected," and "Ignition Enabled.", is powered from an external rechargeable battery to ensure independence from vehicle power fluctuations.

Software Implementation

Development was done in the Arduino IDE. The ESP32 in the Helmet Unit constantly samples sensor data and evaluates ride-readiness through conditional logic:

IF (Helmet worn = TRUE) AND (Alcohol level < Threshold) THEN

Transmit SAFE status → Allow ignition

Else

Transmit UNSAFE status → Inhibit ignition

At the same time, the MPU-6050 monitors acceleration and tilt. If crash thresholds are surpassed, the device provides an SOS alert to allow transmission of location information through GPS data.

Security between the units relies on ESP-NOW, a low-latency, encrypted signalling that discourages any unauthorized system circumvention.

Testing & Validation

The integrated system was evaluated under real-world conditions:

- Helmet worn, no alcohol detected → ignition allowed
- Alcohol detected → ignition blocked
- Helmet not worn → ignition blocked
- Simulated sudden impact → crash alert is triggered Calibration efforts addressed: Tuning of the MQ-3 for a dependable detection in various environmental conditions FSR needs calibration to accurately distinguish real wear from artificial pressure.
- Definition of exact motion thresholds on MPU-6050 to avoid false alarms Overall, the implementation has therefore been able to validate the intended safety logics through ignition control, helmet enforcement, and automated emergency support into an integrated smart system.

CHAPTER 6: RESULT AND DISCUSSION



Fig. 5.1 Helmet unit

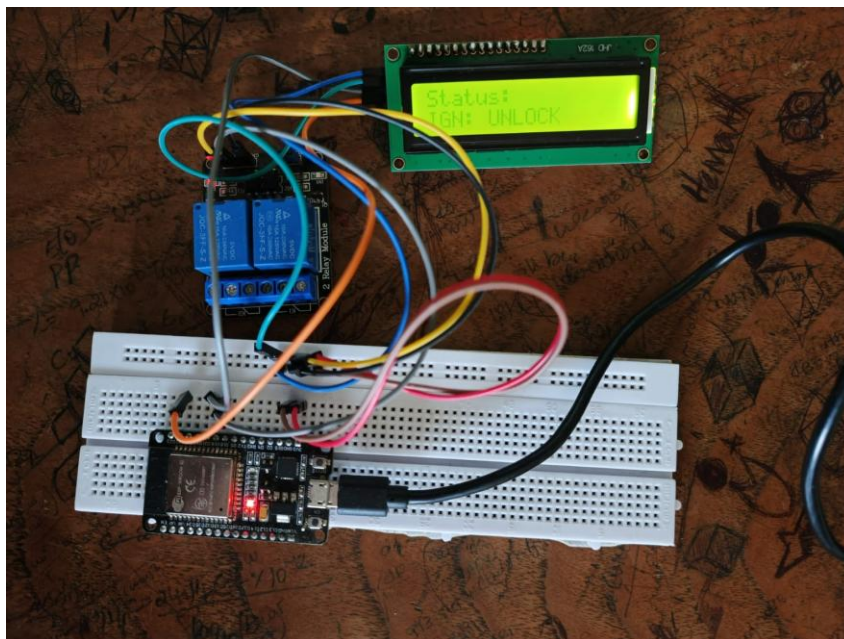


Fig .5.2 Bike unit

The prototype development involved assembling the Helmet Unit and Bike Unit and testing the relay logic.

Ignition Control Results: The system successfully enforces the safety logic. As designed, the bike ignition is enabled *only* when the rider is sober and wearing the helmet.

- **Scenario 1:** Rider puts on the helmet (IR High) and is sober (MQ-3 Low). The system transmits '1', and the relay enables the ignition.
- **Scenario 2:** Rider is drunk (MQ-3 High). The system transmits '0', and the relay cuts the ignition, preventing the bike from starting.

Accident Detection Results: The MPU-6050 successfully measures sudden rotational and linear forces. Upon simulating a crash event (impact trigger), the system identified the deviation and triggered the automated SOS alert logic.

Implementation Challenges: One of the main implementation challenges that were faced involved the calibration of the MQ-3 alcohol sensor. The output signal from this sensor is quite sensitive to ambient conditions, such as temperature and humidity, which caused incongruent readings at first.

This was mitigated by taking the MQ-3 through a proper preheating cycle and stabilizing it in fresh air before every test cycle. The sensor was kept powered on for around 30 seconds and in fresh air, so that the baseline value stabilized at a very consistent reference level. This calibration reduced measurement noise significantly and contributed to robust alcohol detection at runtime.

CHAPTER 7: CONCLUSION AND FUTURE ENHANCEMENT

6.1 Conclusion:

The LifeLine Helmet provides a practical, low-cost, and effective technological solution to improve rider safety. This system tackles two major causes of accidents: drunk driving and helmet non-compliance, by limiting ignition only in cases of sobriety and helmet wear. Safety is enhanced by incorporating automated accident detection, enabling rapid emergency response.

6.2 Future Enhancements:

The project is scalable with several potential upgrades:

1. **AI-Powered Safety:** Implementing AI-powered predictive safety alerts based on rider behavior.
2. **Connectivity:** Integration with 5G IoT for faster communication and location tracking.
3. **Energy Efficiency:** Adding integrated solar charging capabilities to the helmet to extend battery life.
4. **Health Monitoring:** Potential integration with health monitoring and augmented reality (AR) displays for enhanced rider awareness.
5. **Theft Prevention:** Using the helmet pairing as a key, requiring the authorized helmet to be present to start the bike.

REFERENCES:

- [1] IJRPR, “Smart Helmet with Alcohol Detection and Ignition Control,” 2022.
- [2] ResearchGate, “Smart Helmet with Alcohol Sensing and Bike Authentication for Riders,” 2022.
- [3] IARJSET, “Smart Helmet Review with Accident Detection and Communication,” 2024.
- [4] RJWAVE Journal, “Helmet with Alcohol Detection and GPS/GSM for Road Safety,” 2025.
- [5] IJRTI, “Smart Helmet with Accident Detection and Safety Features,” 2025.

APPENDIX:

Components	Price of 1 No. in Rs.	Nos.	Total in Rs.
ESP32 dev module	320	1	320.00
Esp32 c6	544	1	544.00
16x2 LCD Module	170	1	170.00
Charging battery and battery holder	90	1	90.00
5V Relay	35	1	35.00
Breadboard	53	1	53.00
GPS module	293	1	293.00
Accelerometer and gyroscope MPU6050	162	1	162.00
Charging module TPU6050	14	1	14.00
DC voltage booster	35	1	35.00
Dc motor	16	1	16.00
PCB zero board	24	1	24.00
Force sensitive resistor	250	1	250.00
LiPo battery	127	1	127.00
Alcohol Mq-3 sensor	93	1	93.00

